

Multimodal Emotion Recognition — CREMA-D

Thin wrapper — all logic lives in the `emotion_recognition/` package. Upload the folder to Google Drive and set `PACKAGE_DIR` below.

```
! git clone https://github.com/Paresh140/Multi-Modal-Emotion-
Detection/

fatal: destination path 'Multi-Modal-Emotion-Detection' already exists
and is not an empty directory.

! ls
Multi-Modal-Emotion-Detection

! cd /kaggle/working/Multi-Modal-Emotion-Detection
! ls
Multi-Modal-Emotion-Detection

cd Multi-Modal-Emotion-Detection

%cd /content/Multi-Modal-Emotion-Detection
!git pull origin main

! pwd
/kaggle/working

! ls

# — 2. Install deps


---


import subprocess, sys
subprocess.check_call([sys.executable, '-m', 'pip', 'install', '-q',
    'torch', 'torchaudio', 'transformers==4.49',
    'soundfile', 'scikit-learn', 'pandas', 'numpy',
    'matplotlib', 'seaborn', 'tqdm'])

0

# — 3. Point Python at the package


---


import sys

# *** EDIT THESE TWO PATHS ***
```

```
PACKAGE_DIR = '/kaggle/working/Multi-Modal-Emotion-Detection/'
DATA_ROOT = '/kaggle/input/datasets/ejlok1/cremad/AudioWAV'
OUTPUT_DIR = '/kaggle/working/'
```

```
sys.path.insert(0, PACKAGE_DIR)
```

4. Authenticate with Hugging Face

Make sure you have added `HF_TOKEN` to your Colab Secrets (the  icon on the left sidebar) and granted notebook access.

```
#from huggingface_hub import login
#from google.colab import userdata

#try:
#    hf_token = userdata.get('HF_TOKEN')
#    login(token=hf_token)
#    print("Successfully logged in to Hugging Face!")
#except userdata.SecretNotFoundError:
#    print("Warning: HF_TOKEN secret not found. Please add it in the
#    Secrets tab () on the left.")
```

— 5. Run

```
from config import Config
from main import main

cfg = Config(
    data_root=DATA_ROOT,
    output_dir=OUTPUT_DIR,
    use_text=True,
    batch_size=8,
    grad_accum_steps=8, # was 4
    num_epochs=50, # was 20 – give it room
    use_fp16=True,
    freeze_transformer_layers=8, # int, not 10. – confirmed this is
    what worked
)
```

```
results = main(cfg)
print(f"\nFinal test accuracy : {results['acc']:.4f}")
print(f"Final test macro-F1 : {results['macro_f1']:.4f}")
```

```
2026-05-09 19:01:45.896436: E
external/local_xla/xla/stream_executor/cuda/cuda_fft.cc:467] Unable to
register cuFFT factory: Attempting to register factory for plugin
cuFFT when one has already been registered
WARNING: All log messages before absl::InitializeLog() is called are
written to STDERR
```

```
E0000 00:00:1778353305.918266      710 cuda_dnn.cc:8579] Unable to
register cuDNN factory: Attempting to register factory for plugin
cuDNN when one has already been registered
E0000 00:00:1778353305.925044      710 cuda_blas.cc:1407] Unable to
register cuBLAS factory: Attempting to register factory for plugin
cuBLAS when one has already been registered
W0000 00:00:1778353305.942873      710 computation_placer.cc:177]
computation placer already registered. Please check linkage and avoid
linking the same target more than once.
W0000 00:00:1778353305.942893      710 computation_placer.cc:177]
computation placer already registered. Please check linkage and avoid
linking the same target more than once.
W0000 00:00:1778353305.942896      710 computation_placer.cc:177]
computation placer already registered. Please check linkage and avoid
linking the same target more than once.
W0000 00:00:1778353305.942898      710 computation_placer.cc:177]
computation placer already registered. Please check linkage and avoid
linking the same target more than once.
```

```
Device: cuda
GPU: Tesla P100-PCIE-16GB  0.0/17.1 GB used
Loaded 7442 samples | label dist:
emotion
angry      1271
disgust    1271
fear       1271
happy      1271
sad        1271
neutral    1087
```

```
Split → train=5316 (65 speakers) | val=1060 (13 speakers) | test=1066
(13 speakers)
```

```
/usr/local/lib/python3.12/dist-packages/transformers/
configuration_utils.py:315: UserWarning: Passing
`gradient_checkpointing` to a config initialization is deprecated and
will be removed in v5 Transformers. Using
`model.gradient_checkpointing_enable()` instead, or if you are using
the `Trainer` API, pass `gradient_checkpointing=True` in your
`TrainingArguments`.
  warnings.warn(
```

```
AudioEncoder: 12 transformer layers, 8 frozen | trainable=33.5M
frozen=60.9M
```

```
Some weights of RobertaModel were not initialized from the model
checkpoint at roberta-base and are newly initialized:
['roberta.pooler.dense.bias', 'roberta.pooler.dense.weight']
You should probably TRAIN this model on a down-stream task to be able
to use it for predictions and inference.
```

Trainable params: 158.9 M

Starting training...

```
{"model_id":"","version_major":2,"version_minor":0}
```

```
{"model_id":"","version_major":2,"version_minor":0}
```

Ep 01 | train_loss=1.8106 | val_loss=1.6910 | val_acc=0.3500 |
val_mf1=0.3041 ✓ best

```
{"model_id":"","version_major":2,"version_minor":0}
```

```
{"model_id":"","version_major":2,"version_minor":0}
```

Ep 02 | train_loss=1.6008 | val_loss=1.4860 | val_acc=0.4217 |
val_mf1=0.3574 ✓ best

```
{"model_id":"","version_major":2,"version_minor":0}
```

```
{"model_id":"","version_major":2,"version_minor":0}
```

Ep 03 | train_loss=1.3945 | val_loss=1.2848 | val_acc=0.5575 |
val_mf1=0.5441 ✓ best

```
{"model_id":"","version_major":2,"version_minor":0}
```

```
{"model_id":"","version_major":2,"version_minor":0}
```

Ep 04 | train_loss=1.1917 | val_loss=1.1638 | val_acc=0.6519 |
val_mf1=0.6477 ✓ best

```
{"model_id":"","version_major":2,"version_minor":0}
```

```
{"model_id":"","version_major":2,"version_minor":0}
```

Ep 05 | train_loss=1.1038 | val_loss=1.1409 | val_acc=0.6651 |
val_mf1=0.6568 ✓ best

```
{"model_id":"","version_major":2,"version_minor":0}
```

```
{"model_id":"","version_major":2,"version_minor":0}
```

Ep 06 | train_loss=1.0017 | val_loss=1.1722 | val_acc=0.6755 |
val_mf1=0.6650 ✓ best

```
{"model_id":"","version_major":2,"version_minor":0}
```

```
{"model_id":"","version_major":2,"version_minor":0}
```

Ep 07 | train_loss=0.9452 | val_loss=1.0469 | val_acc=0.7000 |
val_mf1=0.6914 ✓ best

```
{"model_id":"","version_major":2,"version_minor":0}
```

```
{"model_id":"","version_major":2,"version_minor":0}

Ep 08 | train_loss=0.9282 | val_loss=1.0263 | val_acc=0.7349 |
val_mf1=0.7318 ✓ best

{"model_id":"","version_major":2,"version_minor":0}

{"model_id":"","version_major":2,"version_minor":0}

Ep 09 | train_loss=0.8525 | val_loss=0.9820 | val_acc=0.7358 |
val_mf1=0.7347 ✓ best

{"model_id":"","version_major":2,"version_minor":0}

{"model_id":"","version_major":2,"version_minor":0}

Ep 10 | train_loss=0.8331 | val_loss=0.9925 | val_acc=0.7396 |
val_mf1=0.7394 ✓ best

{"model_id":"","version_major":2,"version_minor":0}

{"model_id":"","version_major":2,"version_minor":0}

Ep 11 | train_loss=0.8021 | val_loss=1.0609 | val_acc=0.7038 |
val_mf1=0.6962 patience 1/8

{"model_id":"","version_major":2,"version_minor":0}

{"model_id":"","version_major":2,"version_minor":0}

Ep 12 | train_loss=0.7931 | val_loss=1.0319 | val_acc=0.7217 |
val_mf1=0.7177 patience 2/8

{"model_id":"","version_major":2,"version_minor":0}

{"model_id":"","version_major":2,"version_minor":0}

Ep 13 | train_loss=0.7501 | val_loss=1.0109 | val_acc=0.7349 |
val_mf1=0.7332 patience 3/8

{"model_id":"","version_major":2,"version_minor":0}

{"model_id":"","version_major":2,"version_minor":0}

Ep 14 | train_loss=0.7465 | val_loss=1.0924 | val_acc=0.7057 |
val_mf1=0.7004 patience 4/8

{"model_id":"","version_major":2,"version_minor":0}

{"model_id":"","version_major":2,"version_minor":0}

Ep 15 | train_loss=0.7277 | val_loss=1.0149 | val_acc=0.7330 |
val_mf1=0.7315 patience 5/8
```

```
{"model_id":"","version_major":2,"version_minor":0}
{"model_id":"","version_major":2,"version_minor":0}
Ep 16 | train_loss=0.7253 | val_loss=1.0094 | val_acc=0.7349 |
val_mf1=0.7341 patience 6/8
{"model_id":"","version_major":2,"version_minor":0}
{"model_id":"","version_major":2,"version_minor":0}
Ep 17 | train_loss=0.6905 | val_loss=1.0050 | val_acc=0.7415 |
val_mf1=0.7400 ✓ best
{"model_id":"","version_major":2,"version_minor":0}
{"model_id":"","version_major":2,"version_minor":0}
Ep 18 | train_loss=0.6865 | val_loss=1.0618 | val_acc=0.7245 |
val_mf1=0.7214 patience 1/8
{"model_id":"","version_major":2,"version_minor":0}
{"model_id":"","version_major":2,"version_minor":0}
Ep 19 | train_loss=0.6694 | val_loss=1.0298 | val_acc=0.7349 |
val_mf1=0.7326 patience 2/8
{"model_id":"","version_major":2,"version_minor":0}
{"model_id":"","version_major":2,"version_minor":0}
Ep 20 | train_loss=0.6481 | val_loss=1.0065 | val_acc=0.7509 |
val_mf1=0.7495 ✓ best
{"model_id":"","version_major":2,"version_minor":0}
{"model_id":"","version_major":2,"version_minor":0}
Ep 21 | train_loss=0.6552 | val_loss=1.0031 | val_acc=0.7415 |
val_mf1=0.7403 patience 1/8
{"model_id":"","version_major":2,"version_minor":0}
{"model_id":"","version_major":2,"version_minor":0}
Ep 22 | train_loss=0.6359 | val_loss=1.0318 | val_acc=0.7208 |
val_mf1=0.7175 patience 2/8
{"model_id":"","version_major":2,"version_minor":0}
{"model_id":"","version_major":2,"version_minor":0}
```

```
Ep 23 | train_loss=0.6279 | val_loss=0.9993 | val_acc=0.7481 |  
val_mf1=0.7464  patience 3/8
```

```
{"model_id":"","version_major":2,"version_minor":0}
```

```
{"model_id":"","version_major":2,"version_minor":0}
```

```
Ep 24 | train_loss=0.6123 | val_loss=1.0297 | val_acc=0.7368 |  
val_mf1=0.7346  patience 4/8
```

```
{"model_id":"","version_major":2,"version_minor":0}
```

```
{"model_id":"","version_major":2,"version_minor":0}
```

```
Ep 25 | train_loss=0.5994 | val_loss=1.0389 | val_acc=0.7264 |  
val_mf1=0.7220  patience 5/8
```

```
{"model_id":"","version_major":2,"version_minor":0}
```

```
{"model_id":"","version_major":2,"version_minor":0}
```

```
Ep 26 | train_loss=0.5925 | val_loss=1.0508 | val_acc=0.7255 |  
val_mf1=0.7230  patience 6/8
```

```
{"model_id":"","version_major":2,"version_minor":0}
```

```
{"model_id":"","version_major":2,"version_minor":0}
```

```
Ep 27 | train_loss=0.5766 | val_loss=1.0706 | val_acc=0.7396 |  
val_mf1=0.7368  patience 7/8
```

```
{"model_id":"","version_major":2,"version_minor":0}
```

```
{"model_id":"","version_major":2,"version_minor":0}
```

```
Ep 28 | train_loss=0.5747 | val_loss=1.0326 | val_acc=0.7349 |  
val_mf1=0.7319  patience 8/8  
Early stopping at epoch 28
```

```
=== Test Set ===
```

```
/kaggle/working/Multi-Modal-Emotion-Detection/trainer.py:168:  
FutureWarning: You are using `torch.load` with `weights_only=False`  
(the current default value), which uses the default pickle module  
implicitly. It is possible to construct malicious pickle data which  
will execute arbitrary code during unpickling (See  
https://github.com/pytorch/pytorch/blob/main/SECURITY.md#untrusted-models  
for more details). In a future release, the default value for  
`weights_only` will be flipped to `True`. This limits the functions  
that could be executed during unpickling. Arbitrary objects will no  
longer be allowed to be loaded via this mode unless they are  
explicitly allowlisted by the user via  
`torch.serialization.add_safe_globals`. We recommend you start setting
```

`weights\_only=True` for any use case where you don't have full control of the loaded file. Please open an issue on GitHub for any issues related to this experimental feature.

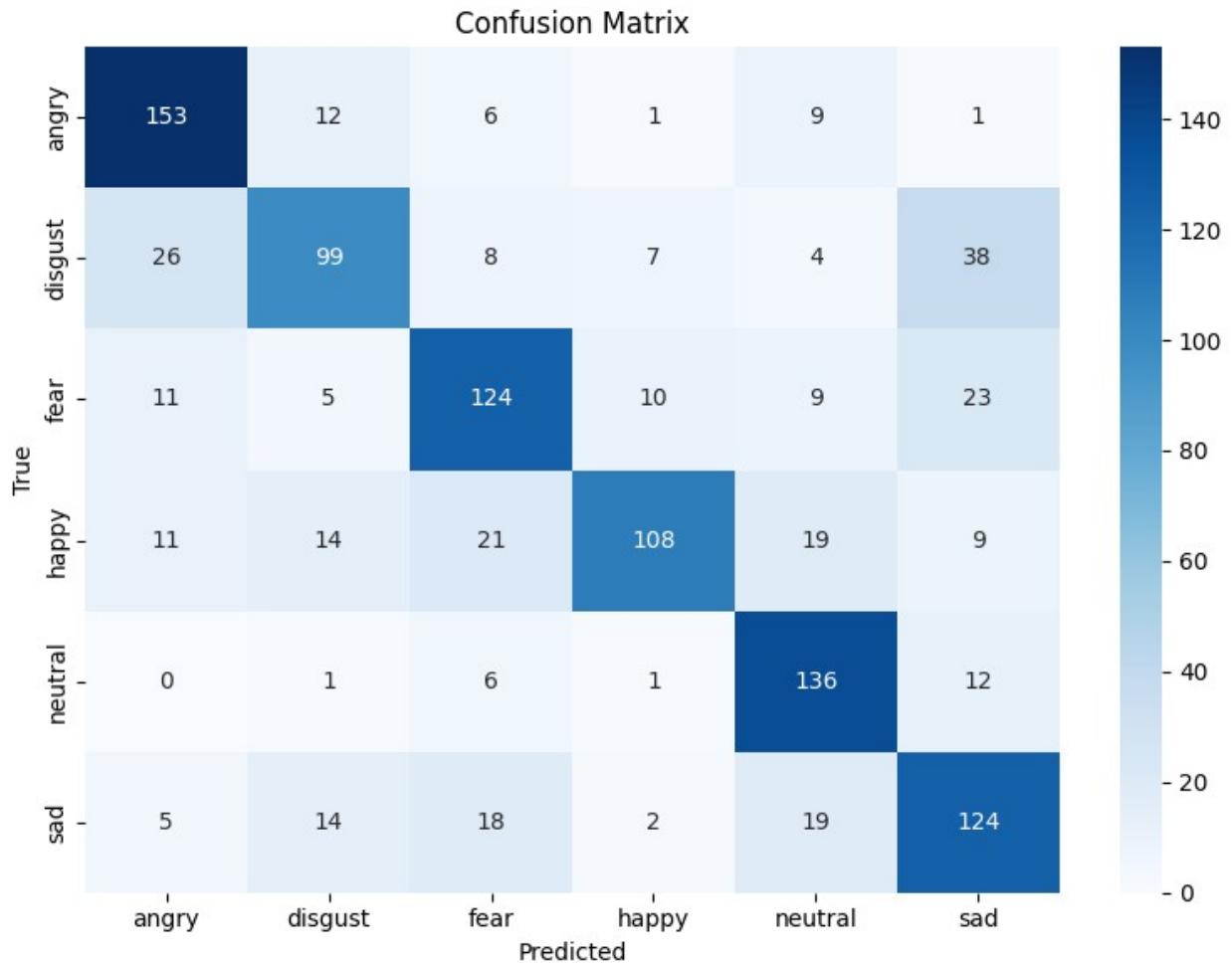
```
self.model.load_state_dict(torch.load(path,
map_location=self.device))
```

```
{"model_id":"","version_major":2,"version_minor":0}
```

Test acc=0.6979 macro_f1=0.6964

=====
Test Report
=====

	precision	recall	f1-score	support
angry	0.7427	0.8407	0.7887	182
disgust	0.6828	0.5440	0.6055	182
fear	0.6776	0.6813	0.6795	182
happy	0.8372	0.5934	0.6945	182
neutral	0.6939	0.8718	0.7727	156
sad	0.5990	0.6813	0.6375	182
accuracy			0.6979	1066
macro avg	0.7055	0.7021	0.6964	1066
weighted avg	0.7058	0.6979	0.6945	1066



Final test accuracy : 0.6979

Final test macro-F1 : 0.6964

! ls

!pip install --force-reinstall torch torchvision torchaudio --index-url https://download.pytorch.org/whl/cu121

! pip install --upgrade bitsandbytes

!pip install --upgrade torch --index-url https://download.pytorch.org/whl/cu124

```
import torch
print(f"GPU: {torch.cuda.get_device_name(0)}")
print(f"PyTorch Version: {torch.__version__}")
print(f"Supported Architectures: {torch.cuda.get_arch_list()}")
```

```
# Force-reinstall a stable version that includes sm_60 support  
!pip install --force-reinstall torch torchvision torchaudio --index-  
url https://download.pytorch.org/whl/cu121  
  
pip install transformers==4.49
```